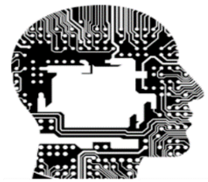
Université Constantine 2
جامعة قسنطينة 2



Apprentissage par Machine 02

APM2

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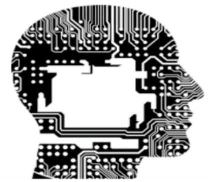

Apprentissage par Machine APM 02
Workshop 01

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Etudiants concernés

Faculté/Institut	Département	Niveau	Spécialité
New Technologies	IFA	Master 01	STIC

Measuring Model Performance

- **Workshop 01**
  - In this exercise, students will apply a supervised machine learning algorithm to the Iris dataset, train a classification model, and evaluate its performance
  - The **Iris** dataset can be loaded from sklearn.datasets
  - Display the first five rows to understand its structure
  - A well-known dataset in machine learning and statistics, commonly used for classification tasks
  - It consists of **150** samples from three species of the Iris flower:
    - Iris Setosa, Iris Versicolor, and Iris Virginica
  - Each sample has four numerical features:
    - Sepal length (cm), Sepal width (cm), Petal length (cm), and Petal width (cm)
  - **Goal 01**: Classify an Iris flower into one of the three species based on these measurements Using KNN and Decision Trees
  - **Goal 02**: Measure and compare the model performance of these classifiers

Measuring Model Performance

- **Workshop 01**
  - In this exercise, students will apply a supervised machine learning algorithm to the Iris dataset, train a classification model, and evaluate its performance
- **STEP 01 : Load the dataset**
  - Load the **Iris** dataset from sklearn.datasets
  - Display the first five rows to understand its structure

ModelPerformance.py

ModelPerformance.py > ...

```python

from sklearn import datasets
import pandas as pd

iris = datasets.load_iris()
X = iris.data
y = iris.target

df = pd.DataFrame(X, columns=iris.feature_names)
df['Species'] = y
print(df.head())
```

Measuring Model Performance

- Workshop 01
  - In this exercise, students will apply a supervised machine learning algorithm to the Iris dataset, train a classification model, and evaluate its performance
- STEP 02: Data Preprocessing
  - Split the dataset into training (80%) and testing (20%) sets
  - Standardize the features using MinMaxScaler or StandardScaler

ModelPerformance.py

ModelPerformance.py > ...

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Measuring Model Performance

- ## Workshop 01
  - In this exercise, students will apply a supervised machine learning algorithm to the Iris dataset, train a classification model, and evaluate its performance
- ## STEP 03: Initialize and Train the Model
  - Select a classification algorithm (KNN and Decision Tree)
  - Initialize and Train each model using the training set

ModelPerformance.py

ModelPerformance.py > ...

```python
from sklearn.neighbors import KNeighborsClassifier

knn_model = KNeighborsClassifier(n_neighbors=5)

knn_model.fit(X_train, y_train)
```

Measuring Model Performance

- Workshop 01
- STEP 04: Model Evaluation
  - Predict the test set labels
  - Compute the accuracy, precision, recall and F1-score
  - Visualize the confusion matrix to analyze model performance

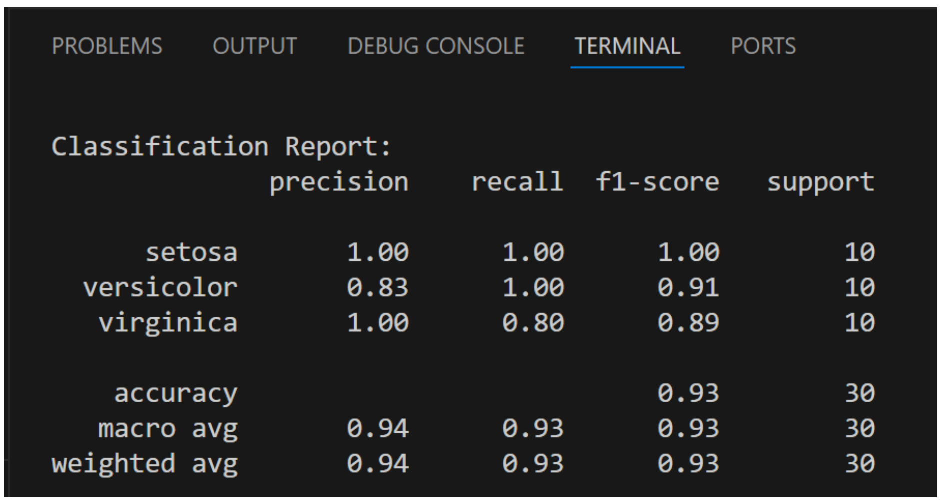
ModelPerformance.py

ModelPerformance.py > ...

```python
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

y_pred = knn_model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy:.2f}")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))

plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt="d", cmap="Blues",
            xticklabels=iris.target_names, yticklabels=iris.target_names)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

Measuring Model Performance

- Workshop 01
- STEP 05: Performance Interpretation
  - Interpret the classification results (for both KNN and Decision Tree) based on the computed evaluation metrics
  - Discuss strengths and weaknesses of each classifier




```
Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       0.83      1.00      0.91        10
   virginica       1.00      0.80      0.89        10

    accuracy                           0.93        30
   macro avg       0.94      0.93      0.93        30
weighted avg       0.94      0.93      0.93        30
```